FA3DHeat: A Frequency-Aware 3-D Heat Conduction Network for Hyperspectral Image Classification

Dong Shi, Zhikang Xiang, *Member, IEEE*, Yu Shen, *Member, IEEE*, Liang Xiao, *Senior Member, IEEE*, and Qian Du, *Fellow, IEEE*

***Abstract*—Hyperspectral image (HSI) classification requires effective spectral–spatial representation learning, since spectral signatures and spatial structures are highly coupled. Although convolutional and attention-based methods have achieved promising performance, efficient modeling of spectral–spatial information in the intrinsic HSI cube remains challenging. Heat conduction-based modeling has recently emerged as a promising approach for efficient long-range information propagation in the frequency domain. However, existing heat conduction-based models are mainly designed for 2-D spatial features and cannot fully characterize the spectral–spatial structure of HSIs. To address this issue, we propose FA3DHeat, a frequency-aware 3-D heat conduction network for HSI classification. Specifically, a 3-D heat conduction operator (3DHCO) is first developed to extend the operation from the 2-D spatial domain to the 3-D spectral–spatial domain. To further exploit frequency-specific information, a multibranch frequency-aware (MBFA) module is designed to extract low-, mid-, and high-frequency representations with learnable soft masks (LSMs). Finally, an adaptive frequency branch fusion (AFBF) module is introduced to dynamically integrate these complementary branches with input-dependent weights. Extensive experiments on three public HSI classification datasets demonstrate that FA3DHeat achieves competitive performance compared with several state-of-the-art methods, validating the effectiveness of 3-D heat conduction for hyperspectral spectral–spatial representation learning. The code will be made publicly available at https://github.com/shidong66014/FA3DHeat**

***Index Terms*—3-D heat conduction, adaptive feature fusion, frequency-aware modeling, hyperspectral image (HSI) classification, spectral–spatial representation learning.**

I. Introduction

HYPERSPECTRAL images (HSIs) acquire detailed spectral signatures across hundreds of contiguous bands, enabling precise characterization of materials. In contrast to multispectral or RGB imagery, HSIs provide richer spectral information and facilitate fine-grained discrimination of surface materials by capturing spectral differences. Benefiting from this advantage, HSIs have been extensively utilized in a wide range of remote sensing applications, including environmental monitoring [1], precision agriculture [2], and mineral exploration [3].

These applications usually require accurate identification of land-cover categories and material types from hyperspectral observations. Therefore, HSI classification has become an essential task for effectively exploiting the information of HSIs in practical scenarios. However, accurate HSI classification remains challenging due to high spectral dimensionality and complex spectral–spatial correlations. To address these challenges, early studies mainly relied on traditional machine learning methods with handcrafted features. Typical methods include support vector machines (SVMs) [4], random forests [5], and Bayesian models [6].

With the rapid development of deep learning, a variety of methods have been increasingly introduced for HSI classification [7]. According to the modeling mechanism, recent methods can be divided into three categories: local receptive field-based methods, attention-based methods, and frequency-domain-based methods [8]. Local receptive field-based methods are mainly represented by convolutional neural networks (CNNs), which first employed 2-D convolutions to extract local spatial features from HSI patches [9]. To better exploit the joint spectral–spatial information of HSI cubes, 3-D CNNs and hybrid 2-D–3-D CNNs are further developed [10], [11], where convolutional operations are performed along both spectral and spatial dimensions, offering promising classification performance. Attention-based methods, especially Transformer-based models, can capture long-range dependencies through self-attention mechanisms. To effectively model spectral–spatial features, several representative transformer-based methods have been developed. For example, SSFTT introduces spectral–spatial feature tokenization to capture both low-level spectral–spatial features and high-level semantic representations [12]. MSTNet further designs a multilevel spectral–spatial Transformer architecture to enhance hierarchical feature representation [13]. More recently, state-space models, such as Mamba, have been introduced into HSI analysis as an

Received 1 July 2026; revised 25 August 2026; accepted 1 September 2026. Date of publication 9 September 2026; date of current version 18 September 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62501286. *(Corresponding author: Yu Shen.)*

Dong Shi and Yu Shen are with the College of Information Science and Technology & Artificial Intelligence, Nanjing Forestry University, Nanjing 210037, China (e-mail: shenyu01@njfu.edu.cn).

Zhikang Xiang is with the Department of Computer Information and Network Security, Jiangsu Police Institute, Nanjing 210031, China.

Liang Xiao is with the School of Computer Science and Engineering, Nanjing University of Science and Technology, Nanjing 210094, China.

Qian Du is with the Department of Electrical and Computer Engineering, Mississippi State University, Starkville, MS 39762 USA.

Digital Object Identifier 10.1109/TGRS.2026.3732130

1558-0644 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:20:38 UTC from IEEE Xplore. Restrictions apply.

efficient alternative. By employing selective state-space transitions, Mamba can capture long-range dependencies with linear computational complexity, providing an effective solution for efficient spectral–spatial modeling [14], [15], [16]. Despite these advantages, existing Mamba-based HSI classification methods usually transform spatial or spectral–spatial features into sequential representations for scanning. This sequential modeling strategy may disrupt the intrinsic spatial structure and local continuity among neighboring pixels. Frequency domain-based methods aim to enhance HSI representation by exploiting frequency information contained in spectral or spectral–spatial features. For example, Fourier-based methods enhance spectral separability by analyzing spectral signatures in the frequency domain, while wavelet-based methods decompose HSI features into multiscale frequency components for discriminative representation learning [17], [18].

More recently, heat conduction has emerged as a physics-inspired model for frequency-domain visual representation learning [19]. Unlike conventional frequency-domain methods that mainly focus on feature decomposition or enhancement, heat conduction formulates feature interaction as an anisotropic diffusion-based propagation process. By solving the diffusion equation in the frequency domain, it enables efficient frequency-wise information propagation without the dense pairwise computation required by self-attention. This propagation mechanism has shown promising performance in general vision tasks, such as image classification and semantic segmentation [20]. However, existing visual heat conduction models were originally developed for 2-D images and mainly formulate heat diffusion over the spatial dimensions. Although HCI-Net introduces heat conduction into HSI classification [21], it handles spectral information through separate feature extraction, leaving its heat conduction operator limited to 2-D spatial diffusion. Such a design is insufficient to fully capture the coupled spectral–spatial dependencies in HSIs, where spectral bands and spatial structures jointly determine the semantic representation. *Therefore, how to effectively leverage heat conduction for the 3-D spectral–spatial representations of HSIs remains challenging.*

To address this limitation, we extend heat conduction from the 2-D spatial domain to the 3-D spectral–spatial domain and propose FA3DHeat, a frequency-aware 3-D heat conduction network for HSI classification. The key idea of FA3DHeat is to introduce a 3-D heat conduction operator (3DHCO), which regards hyperspectral features as a 3-D conduction field, where information propagation is performed jointly along the spectral and spatial dimensions. As illustrated in Fig. 1(a), conventional frequency-domain processing often decomposes or enhances features and applies frequency responses that are predefined or uniformly applied across directions for subsequent filtering, reweighting, or fusion, without explicitly modeling a direction-dependent diffusion process. Fig. 1(b) further shows that heat conduction likewise operates in the frequency domain but derives its propagation response from a diffusion equation, enabling anisotropic heat diffusion for global feature propagation; however, this diffusion remains limited to a 2-D spatial field. Building on this diffusion mechanism, Fig. 1(c) shows that the proposed 3DHCO treats the spectral

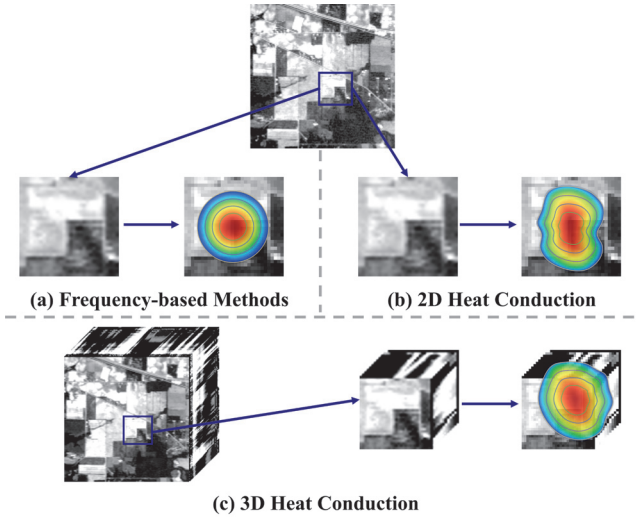


Fig. 1. Comparison of (a) frequency-based feature processing, (b) 2-D spatial heat conduction, and (c) 3-D spectral–spatial heat conduction. Compared with 2-D processing, 3-D heat conduction performs joint diffusion over the HSI cube along both spatial and spectral dimensions.

dimension as the third conduction coordinate and extends anisotropic heat diffusion to the 3-D spectral–spatial field, enabling frequency-aware information propagation across both spectral and spatial dimensions through 3-D-discrete cosine transform (DCT)/IDCT and direction-specific diffusion coefficients. Furthermore, a multibranch frequency-aware (MBFA) module is proposed to exploit the complementary information contained in different frequency components. Specifically, multiple parallel 3DHCO branches are constructed to aggregate low-, mid-, and high-frequency representations, thereby enhancing the discriminative spectral–spatial representation of HSIs. Finally, an adaptive frequency branch fusion (AFBF) module is further designed to dynamically integrate these frequency-specific representations. In this way, FA3DHeat can better match the intrinsic structure of HSI cubes and learn more discriminative spectral–spatial representations. The main contributions of this work are summarized as follows.

1) A frequency-aware 3-D heat conduction network, named FA3DHeat, is proposed for HSI classification. In this network, a 3-D heat conduction operator (3DHCO) is introduced to extend heat conduction modeling from the 2-D spatial domain to the 3-D spectral–spatial domain. By treating hyperspectral features as a 3-D conduction field, 3DHCO enables efficient information propagation along both spectral and spatial dimensions.
2) An MBFA module is designed to capture complementary spectral–spatial representations from different frequency components. Meanwhile, an AFBF module is introduced to dynamically integrate these frequency-specific features according to input-dependent spectral–spatial statistics.
3) Extensive experiments conducted on three public HSI datasets demonstrate that the proposed FA3DHeat achieves competitive performance compared with several state-of-the-art methods, validating the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:20:38 UTC from IEEE Xplore. Restrictions apply.

effectiveness of frequency-aware 3-D heat conduction for spectral–spatial representation learning.

The remainder of this article is organized as follows. Section II reviews the related work. Section III details the proposed network. Section IV describes the datasets and experimental setup, followed by the experimental results and analysis. Finally, Section V concludes this study.

II. Related Work

A. Deep Learning-Based HSI Classification

Deep learning-based HSI classification methods can be broadly categorized into local receptive field-based methods, attention-based methods, and frequency-based methods. Local receptive field-based methods, mainly represented by CNN-based models, have been widely studied due to their strong capability in local feature extraction. Early CNN-based methods used 2-D convolutions to capture spatial patterns or 1-D convolutions to model spectral responses, while later methods introduced 3-D convolutions for joint spectral–spatial representation learning. Hybrid 2-D and 3-D CNN convolutional networks have also been developed to exploit complementary information from different dimensions [22], [23], [24]. Recent studies further employ multiscale convolution, large-kernel convolution, and spatial mapping strategies to enlarge the receptive field and improve contextual modeling [25], [26], [27]. Although these methods achieve promising performance, their local receptive fields still limit the modeling of long-range dependencies and global contextual relationships [28]. Attention-based methods have been introduced to address this limitation. Transformer-based methods treat spectral bands, spatial patches, or spectral–spatial features as token sequences and use self-attention to capture global contextual information. Representative methods include SpectralFormer, which learns group-wise spectral embeddings for neighboring spectral bands [29], and SSFTT, which introduces spectral–spatial feature tokenization for compact semantic representation [12]. Hierarchical and lightweight Transformer variants have also been developed to improve spectral–spatial interaction and reduce computational cost [30], [31], [32], [33], [34], [35]. More recently, state-space models, especially Mamba, have been explored for efficient long-range dependency modeling with linear complexity. Typical examples include MambaHSI [36], HyperMamba [14], IGroupSS-Mamba [37], GraphMamba [16], and HyPyra-Mamba [38]. However, Transformer-based methods usually suffer from high computational and memory costs, while Mamba-based methods rely on sequential scanning strategies that may not fully preserve the intrinsic spatial continuity of HSI data.

Frequency-based methods provide a complementary perspective for HSI classification by representing global structures and local details with frequency components [17], [39]. Based on this, several studies have introduced frequency-domain transformations, such as fast Fourier transform (FFT), DCT, and wavelet-based decomposition, to enhance spectral–spatial representation learning. For example, dual-frequency Transformer frameworks have been developed to jointly exploit low- and high-frequency information for HSI classification [40]. Frequency filtering and split-frequency strategies have also been explored to strengthen feature interaction and preserve detailed information in the frequency domain [41], [42]. Although these methods demonstrate the effectiveness of frequency-domain modeling, most of them mainly focus on 2-D frequency transformation, and 3-D spectral–spatial feature representation remains insufficiently explored.

B. Heat Conduction-Based Visual Representation

Recently, heat conduction-based modeling has been explored in computer vision for effective information propagation and feature aggregation. Compared with conventional frequency-based methods, which often perform frequency decomposition or filtering in a relatively uniform manner, heat conduction provides an anisotropic propagation mechanism in the frequency domain. It enables direction-dependent modulation of feature responses, allowing the model to adaptively control information propagation along different dimensions. For example, Wang et al. [19] developed a representative heat conduction-based visual representation model, where image patches are regarded as heat sources, and their correlations are modeled through thermal energy diffusion, achieving effective global receptive field modeling. Following this idea, heat conduction mechanisms have been applied to general visual representation learning and remote sensing tasks, showing promising potential in image classification, object detection, and semantic segmentation [20], [43]. For HSI classification, Liang et al. [21] incorporated heat conduction into spectral–spatial feature modeling through HCI-Net; however, its heat conduction operator defines heat diffusion only over the spatial dimensions, while spectral information is handled separately outside the diffusion equation.

Thus, although HCI-Net learns spectral–spatial representations at the network level, its heat equation contains neither an explicit spectral diffusion term nor a spectral-direction diffusivity, and the operator itself remains limited to 2-D spatial diffusion. Therefore, in this article, we extend heat conduction from the 2-D spatial domain to the 3-D spectral–spatial domain and propose FA3DHeat, a frequency-aware 3-D heat conduction network for HSI classification. By designing 3DHCO, the proposed method performs anisotropic information propagation along both spectral and spatial dimensions, leading to more discriminative spectral–spatial representations.

III. Proposed Method

In this section, we first introduce the preliminaries of heat conduction, and then the proposed FA3DHeat is described in detail.

A. Preliminaries of Heat Conduction

Let $v(x, y, t)$ denote the temperature at spatial position (x, y) and time t in a 2-D continuous spatial domain. According to classical heat conduction theory [44], the heat diffusion process is governed by

$$\frac{\partial v}{\partial t} = k\left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2}\right) \tag{1}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:20:38 UTC from IEEE Xplore. Restrictions apply.

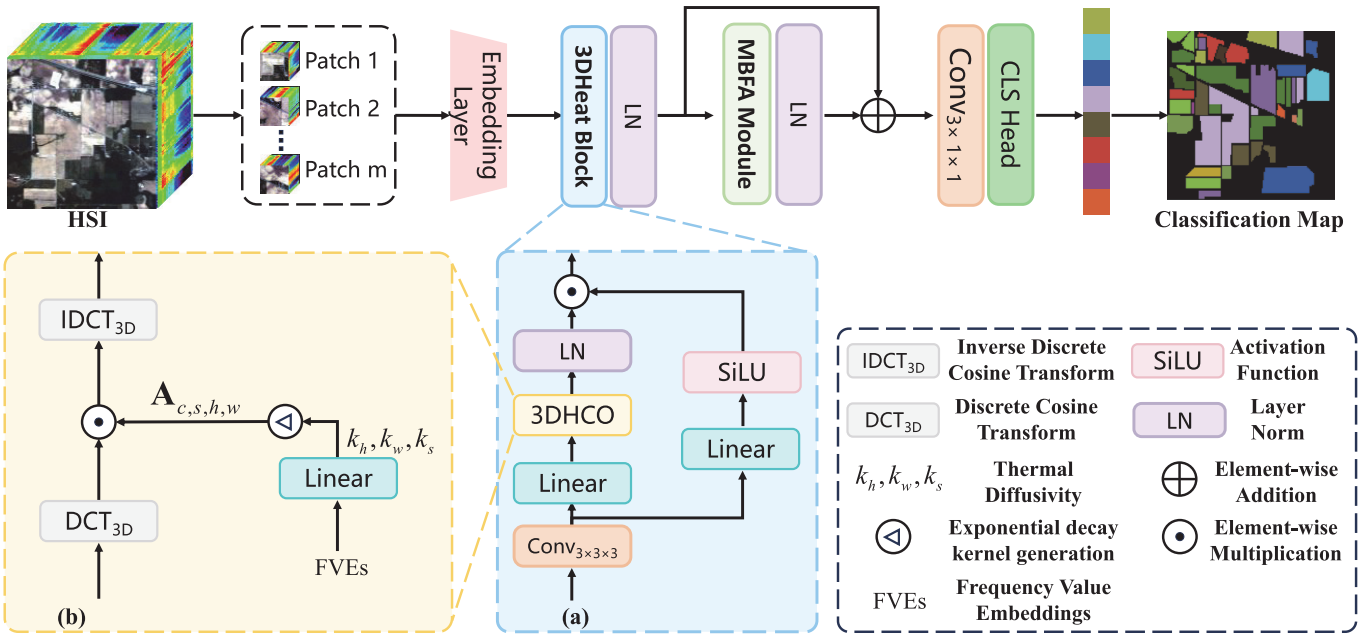


Fig. 2. Overall architecture of the proposed FA3DHeat. The network first partitions the input HSI into local patches, followed by a 3DHeat block and an MBFA module for frequency-aware spectral–spatial representation learning. (a) 3DHeat block. (b) 3-D heat conduction operator.

where $k > 0$ is the thermal diffusivity coefficient.

Given the initial condition $v(x, y, 0) = g(x, y)$, the solution of (1) can be derived in the frequency domain. Let $\tilde{v}(\omega_x, \omega_y, t) = \mathcal{F}_{x,y}(v(x, y, t))$ and $\tilde{g}(\omega_x, \omega_y) = \mathcal{F}_{x,y}(g(x, y))$, where ω_x and ω_y denote spatial frequencies. Using the derivative property of the Fourier transform, (1) is transformed into

$$\frac{\partial \tilde{v}\left(\omega_x, \omega_y, t\right)}{\partial t} = -k\left(\omega_x^2 + \omega_y^2\right)\tilde{v}\left(\omega_x, \omega_y, t\right). \tag{2}$$

Solving this ordinary differential equation with the initial condition $\tilde{v}(\omega_x, \omega_y, 0) = \tilde{g}(\omega_x, \omega_y)$ gives

$$\tilde{v}\left(\omega_x, \omega_y, t\right) = \tilde{g}\left(\omega_x, \omega_y\right) e^{-k(\omega_x^2+\omega_y^2)t}. \tag{3}$$

Finally, the spatial-domain solution can be obtained by the inverse Fourier transform

$$v\left(x, y, t\right) = \mathcal{F}_{x,y}^{-1}\left[\tilde{g}\left(\omega_x, \omega_y\right) e^{-k(\omega_x^2+\omega_y^2)t}\right]. \tag{4}$$

For visual feature modeling, heat conduction-based methods extend k to an adaptive thermal diffusivity by predicting it from frequency value embeddings (FVEs), which play a role similar to absolute position embeddings in ViTs. This adaptive thermal diffusivity improves the flexibility of heat conduction operators and provides a useful basis for constructing frequency-domain feature propagation modules in neural networks.

B. Architecture Overview

The overall architecture of the proposed FA3DHeat is illustrated in Fig. 2. Given an input HSI patch $\mathbf{X} \in \mathbb{R}^{S\times H\times W}$, where S denotes the number of spectral bands and H and W denote the spatial height and width, respectively. A shallow embedding layer, implemented by a learnable spectral reducer, is first adopted to project the original spectral dimension from S to a reduced spectral feature dimension S_r. The embedded feature is then fed into a 3DHeat block, where the proposed 3-D heat conduction operator (3DHCO) is employed to perform frequency-domain 3-D spectral–spatial information propagation.

To further capture complementary frequency information, an MBFA module is applied. The MBFA module contains three parallel low-, mid-, and high-frequency branches for complementary frequency-aware representation learning. These branches produce different heat diffusion responses to emphasize different frequency components. Specifically, low-frequency responses are used to describe global contextual consistency, mid-frequency responses are used to capture local structural variations, and high-frequency responses are used to preserve fine-grained spectral–spatial details. An AFBF module is then developed to dynamically integrate these frequency-specific representations according to input-dependent spectral–spatial statistics. The fused feature is subsequently combined with the input of the MBFA module through a residual connection, and the resulting feature is fed into a classification head to predict the category label of the central pixel.

C. 3-D Heat Conduction

1) 3-D Heat Conduction Operator (3DHCO): The heat conduction formulation introduced above describes diffusion over a 2-D spatial domain. However, an HSI naturally forms a 3-D spectral–spatial volume, where the spectral dimension contains continuous material responses and is strongly coupled with spatial structures. Therefore, directly applying 2-D heat diffusion is insufficient to characterize the intrinsic structure of hyperspectral data. As illustrated in Fig. 1, the heat conduction process is extended from the 2-D spatial domain to the 3-D

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:20:38 UTC from IEEE Xplore. Restrictions apply.

spectral–spatial domain, where information propagation is performed along the spectral, height, and width dimensions.

Let $u(h, w, s, t)$ denote the feature intensity at spectral–spatial position (h, w, s) and propagation time t, where h and w are spatial coordinates and s denotes the spectral coordinate. To model direction-dependent propagation behaviors, the anisotropic 3-D heat conduction process is formulated as

$$\frac{\partial u}{\partial t} = k_h \frac{\partial^2 u}{\partial h^2} + k_w \frac{\partial^2 u}{\partial w^2} + k_s \frac{\partial^2 u}{\partial s^2} \tag{5}$$

where $k_h, k_w, k_s > 0$ are the thermal diffusivity coefficients along the height, width, and spectral dimensions, respectively. This formulation allows different propagation strengths to be assigned to different spectral–spatial directions.

Given the initial condition $u(h, w, s, 0) = f(h, w, s)$, (5) can be solved in the frequency domain. Let $\tilde{u}(\omega_h, \omega_w, \omega_s, t)$ and $\tilde{f}(\omega_h, \omega_w, \omega_s)$ denote the Fourier transforms of $u(h, w, s, t)$ and $f(h, w, s)$, respectively. According to the derivative property of the Fourier transform, (5) is transformed into

$$\frac{\partial \tilde{u}}{\partial t} = -\left(k_h \omega_h^2 + k_w \omega_w^2 + k_s \omega_s^2\right) \tilde{u} \tag{6}$$

where ω_h, ω_w, and ω_s are the frequency coordinates along the corresponding dimensions. The solution is then obtained as

$$\tilde{u}(\omega_h, \omega_w, \omega_s, t) = \tilde{f}(\omega_h, \omega_w, \omega_s) \times \exp\left[-t\left(k_h \omega_h^2 + k_w \omega_w^2 + k_s \omega_s^2\right)\right]. \tag{7}$$

For neural network implementation, the continuous Fourier-domain solution is discretized by using 3D-DCT and 3D-IDCT. Given a hyperspectral feature tensor $\mathbf{F} \in \mathbb{R}^{C \times S_r \times H \times W}$ obtained after local 3-D convolution and feature projection, where C denotes the learned feature-channel dimension, 3D-DCT is applied independently to each feature channel along the reduced spectral and spatial dimensions. The proposed 3DHCO is defined as

$$\mathbf{X_{3DHCO}} = \mathcal{D}_{3D}^{-1}\left[\mathcal{D}_{3D}(\mathbf{F}) \odot \mathbf{A}\right] \tag{8}$$

where $\mathcal{D}_{3D}(\cdot)$ and $\mathcal{D}_{3D}^{-1}(\cdot)$ denote the 3D-DCT and 3D-IDCT operators, respectively, and $\odot$ represents element-wise multiplication. The heat attenuation term $\mathbf{A}$ is formulated as

$$\mathbf{A}_{c,s,h,w} = \exp\left(-k_{s,c}\omega_s^2 - k_{h,c}\omega_h^2 - k_{w,c}\omega_w^2\right) \tag{9}$$

where $k_{s,c}$, $k_{h,c}$, and $k_{w,c}$ are channel-wise diffusion coefficients, and ω_s, ω_h, and ω_w denote the discrete frequency coordinates along the spectral, height, and width dimensions. In practice, the propagation time t is absorbed into the learnable diffusion coefficients, and the positivity of these coefficients can be ensured by a nonnegative activation function.

Although (9) introduces dimension-dependent diffusion coefficients, these coefficients are shared across all frequency positions within each channel. To further capture frequency-position-specific characteristics, we introduce FVEs for 3D-DCT coefficients. Each FVE encodes the position of a spectral–spatial frequency component and is projected to adjust the diffusion coefficients along the spectral, height, and width dimensions. In this way, the heat decay kernel becomes both direction-dependent and frequency-position-adaptive, enabling more flexible anisotropic spectral–spatial propagation. The details are illustrated in Fig. 2(b).

2) 3DHeat Block: Based on the proposed 3DHCO, the 3DHeat block is constructed to integrate local spectral–spatial interaction and frequency-domain heat diffusion, as shown in Fig. 2(a). Given an input feature tensor $\mathbf{X} \in \mathbb{R}^{C \times S_r \times H \times W}$, a local 3-D convolution is first applied to introduce neighborhood-aware spectral–spatial context

$$\mathbf{X}_l = \mathrm{Conv}_{3\times3\times3}(\mathbf{X}). \tag{10}$$

Then, a pointwise projection is used to generate two branches

$$(\mathbf{F}, \mathbf{Z}) = \mathrm{Split}\left(\mathrm{Proj}_{in}(\mathbf{X}_l)\right) \tag{11}$$

where $\mathbf{F}$ is fed into the heat conduction branch, and $\mathbf{Z}$ is used as a gating branch for feature modulation. The heat conduction branch applies the previously defined 3DHCO to $\mathbf{F}$ and produces the propagated feature $\mathbf{X_{3DHCO}}$, enabling efficient frequency-domain spectral–spatial propagation.

After heat conduction, gated feature modulation and output projection are performed to obtain the block output

$$\mathbf{X}_{out} = \mathrm{Proj}_{out}\left(\mathrm{LN}(\mathbf{X}_{3\mathrm{DHCO}}) \odot \mathrm{SiLU}(\mathbf{Z})\right) \tag{12}$$

where $\mathrm{LN}(\cdot)$ denotes layer normalization, $\mathrm{SiLU}(\cdot)$ is the activation function, and $\odot$ represents element-wise multiplication.

In this block, the local 3-D convolution captures short-range spectral–spatial patterns, while 3DHCO provides frequency-domain long-range propagation over the hyperspectral feature volume. The gating branch further improves feature selectivity, making the 3DHeat Block suitable for efficient spectral–spatial representation learning.

D. MBFA Module

1) Frequency-Aware Branch Construction: After the preceding 3DHeat Block, the output feature $\mathbf{X}_{out}$ has encoded both local spectral–spatial interactions and long-range heat-based propagation. To further exploit complementary frequency information, the MBFA module is introduced to construct low-, mid-, and high-frequency branches based on $\mathbf{X}_{out}$. By separately modeling low-, mid-, and high-frequency components, these branches capture complementary global and local spectral–spatial representations. As illustrated in Fig. 3, the MBFA module consists of three parallel frequency-aware branches, each implemented with an independently parameterized 3DHeat Block, followed by an AFBF unit for adaptive branch aggregation. Within each branch, the corresponding mask is applied inside 3DHCO, where it jointly modulates the frequency-domain feature together with the branch-specific heat attenuation response before 3D-IDCT.

Let $\mathcal{B} = \{l, m, h\}$ denote the low-, mid-, and high-frequency branches, respectively. For the bth branch, the local 3-D convolution and input projection of its 3DHeat Block generate a heat-conduction feature $\mathbf{U}^b$ and a gating feature $\mathbf{Z}^b$. The branch-specific heat-conduction feature is transformed into the frequency domain as

$$\hat{\mathbf{U}}^b = \mathcal{D}_{3D}\left(\mathbf{U}^b\right), \quad b \in \mathcal{B}. \tag{13}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:20:38 UTC from IEEE Xplore. Restrictions apply.

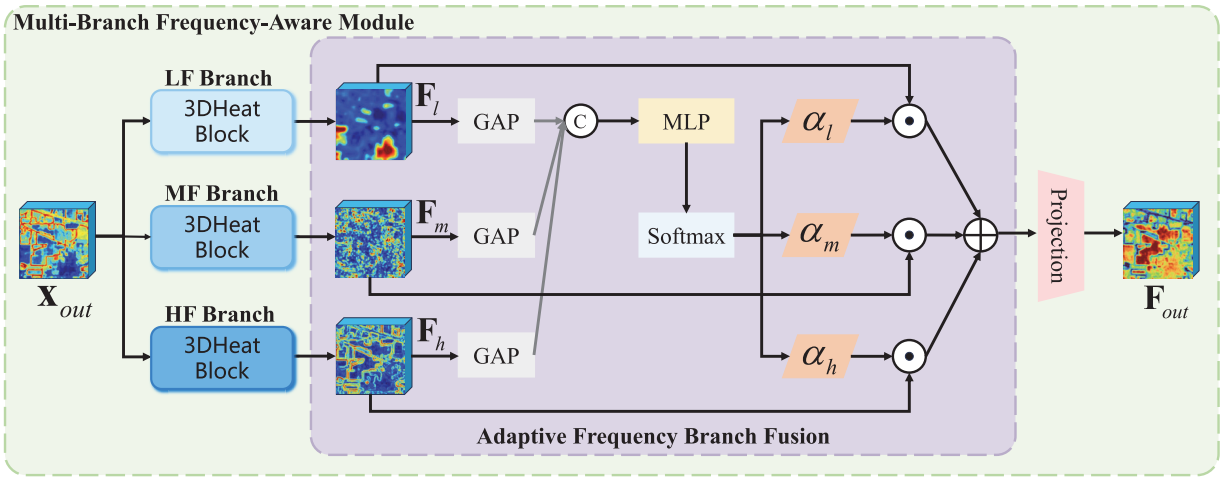


Fig. 3. Illustration of the proposed MBFA module. MBFA extracts low-, mid-, and high-frequency spectral–spatial representations through three parallel 3DHeat Blocks, and adaptively fuses these complementary features using AFBF to generate the refined output $\mathbf{F}_{out}$.

To characterize the frequency level of each 3D-DCT coefficient, we define a normalized spectral–spatial frequency magnitude as

$$\rho_{s,h,w} = \frac{\sqrt{\omega_s^2 + \omega_h^2 + \omega_w^2}}{\max_{s,h,w} \sqrt{\omega_s^2 + \omega_h^2 + \omega_w^2} + \epsilon} \tag{14}$$

where ω_s, ω_h, and ω_w denote the discrete frequency coordinates along the spectral, height, and width dimensions, respectively, and ϵ is a small constant for numerical stability. Based on $\rho_{s,h,w}$, we generate learnable soft masks (LSMs) for the three frequency branches. For the bth branch, the mask score is defined as

$$\mathbf{Q}^b_{s,h,w} = -\frac{\left(\rho_{s,h,w} - \mu_b\right)^2}{2\sigma_b^2 + \epsilon} \tag{15}$$

where μ_b and σ_b denote the learnable frequency center and bandwidth of the bth branch, respectively. The frequency centers are initialized with ordered low-, mid-, and high-frequency priors, and the bandwidths are constrained to be positive during training. In practical implementation, a cumulative Softmax reparameterization is adopted to preserve the ordering of the frequency centers, while Softplus is used to ensure positive bandwidths.

The three mask scores are normalized along the branch dimension to obtain branch-wise frequency masks

$$\mathbf{M}^b_{s,h,w} = \frac{\exp\left(\mathbf{Q}^b_{s,h,w}\right)}{\sum_{j\in\mathcal{B}} \exp\left(\mathbf{Q}^j_{s,h,w}\right)}, \quad b \in \mathcal{B}. \tag{16}$$

This competitive normalization encourages complementary responses among different branches and softly assigns each 3D-DCT coefficient to a suitable frequency component without using manually defined hard thresholds.

Within the 3DHCO of the bth branch, the branch-specific mask $\mathbf{M}^b$ is applied jointly with the corresponding heat attenuation response $\mathbf{A}^b$. Specifically, the propagated feature inside the branch is obtained as

$$\mathbf{Y}^b = \mathcal{D}_{3D}^{-1}\left(\hat{\mathbf{U}}^b \odot \mathbf{A}^b \odot \mathbf{M}^b\right), \quad b \in \mathcal{B} \tag{17}$$

where $\mathbf{A}^b$ is the heat attenuation response generated by the independently parameterized 3DHCO in the bth branch, following (9), and $\mathbf{M}^b$ is broadcast along the batch and channel dimensions.

Following the normalization, gated modulation, and output projection of the corresponding 3DHeat block, the final representation of the bth frequency branch is formulated as:

$$\mathbf{F}_b = \mathrm{Proj}^b_{out}\left(\mathrm{LN}^b\left(\mathbf{Y}^b\right) \odot \mathrm{SiLU}\left(\mathbf{Z}^b\right)\right), \quad b \in \mathcal{B} \tag{18}$$

where b indicates that the 3DHeat block parameters are independently learned for each frequency branch.

2) Adaptive Frequency Branch Fusion: According to (18), the frequency-aware branches produce three complementary representations, denoted as $\mathbf{F}_l$, $\mathbf{F}_m$, and $\mathbf{F}_h$ for the low-, mid-, and high-frequency branches, respectively. These branches emphasize different spectral–spatial properties. A direct summation or concatenation treats these branches with fixed importance, which is not flexible enough for hyperspectral scenes with diverse spectral–spatial variations. Therefore, we introduce an AFBF module to dynamically integrate the three frequency branches.

AFBF first summarizes the global response of each frequency branch by global average pooling over the reduced spectral and spatial dimensions. Given the branch feature $\mathbf{F}_b \in \mathbb{R}^{C\times S_r\times H\times W}$, GAP is performed over the reduced spectral and spatial dimensions

$$\mathbf{g}_b = \mathrm{GAP}\left(\mathbf{F}_b\right), \quad b \in \{l, m, h\} \tag{19}$$

where $\mathbf{g}_b \in \mathbb{R}^C$ is the channel descriptor of the bth frequency branch. The three descriptors are concatenated and fed into a lightweight MLP to model their cross-branch dependencies. Instead of predicting one scalar weight for each branch, the MLP generates channel-wise branch scores

$$[\mathbf{e}_l, \mathbf{e}_m, \mathbf{e}_h] = \mathrm{MLP}\left(\mathrm{Concat}\left[\mathbf{g}_l, \mathbf{g}_m, \mathbf{g}_h\right]\right) \tag{20}$$

where $\mathbf{e}_b \in \mathbb{R}^C$ denotes the channel-wise score of the bth frequency branch.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:20:38 UTC from IEEE Xplore. Restrictions apply.

The branch scores are then normalized along the frequency-branch dimension to obtain channel-wise fusion weights α_l, α_m, and α_h. This normalization adaptively balances the contributions of low-, mid-, and high-frequency branches for each channel. Therefore, different channels can emphasize different frequency responses according to the global spectral–spatial characteristics of the input.

The adaptively fused feature is obtained by channel-wise weighted aggregation followed by a point-wise projection

$$\mathbf{F}_{out} = \mathrm{Proj}\left(\alpha_l \odot \mathbf{F}_l + \alpha_m \odot \mathbf{F}_m + \alpha_h \odot \mathbf{F}_h\right) \tag{21}$$

where α_l, α_m, and α_h are broadcast to match the spectral and spatial dimensions of the corresponding branch features. With this design, AFBF performs input-dependent and channel-wise frequency selection. Compared with fixed fusion strategies, AFBF provides a more flexible way to integrate complementary frequency representations while keeping the module lightweight. Following the entire MBFA module, its input $\mathbf{X}_{out}$ is combined with the normalized adaptively fused feature $\mathbf{F}_{out}$ through a residual connection:

$$\mathbf{X}_{\mathrm{Final}} = \mathbf{X}_{out} + \mathrm{LN}\left(\mathbf{F}_{out}\right). \tag{22}$$

The resulting feature $\mathbf{X}_{\mathrm{Final}}$ is then fed into a lightweight classification head for final classification. During training, the proposed network is optimized in an end-to-end manner using the standard cross-entropy loss between the predicted class probabilities and the ground-truth labels.

IV. Experiment

A. Datasets and Evaluation Metrics

To evaluate the classification performance of the proposed FA3DHeat, three public HSI datasets are used in the experiments, including Indian Pines, Augsburg, and Houston 2013.

1) Indian Pines: The Indian Pines dataset was collected by the airborne visible infrared imaging spectrometer (AVIRIS) [45] sensor over the Indian Pines test site in northwestern Indiana, USA. It consists of 145 × 145 pixels and 224 spectral bands, covering the wavelength range from 0.4 to 2.5 μm with a spectral resolution of 10 nm. After removing noisy and water absorption bands, 200 spectral bands are retained for analysis. The dataset includes 16 land cover classes, such as corn, wheat, grass pasture, and woods. The detailed sample split is reported in Table I.

2) Augsburg: The Augsburg dataset was acquired by the HySpex sensor over the city of Augsburg, Germany [46]. It features a spatial resolution of 30 m with an image size of 332 × 485 pixels. This dataset comprises 180 spectral bands covering the visible and near-infrared spectral range from 400 to 2500 nm. The detailed sample split is reported in Table II.

3) Houston 2013: The Houston 2013 dataset was acquired over the University of Houston campus and its neighboring urban area using the ITRES CASI-1500 hyperspectral sensor [47]. The scene comprises 349 × 1905 pixels with a spatial resolution of 2.5 m. It contains 144 spectral channels spanning the wavelength range of 380–1050 nm at a spectral sampling interval of approximately 5 nm. The detailed sample split is reported in Table III.

TABLE I
Land-Cover Categories of the Indian Pines Dataset and Its Corresponding Training and Testing Samples

No.	Category	Training	Testing	Color
1	Corn Notill	50	1384	
2	Corn Mintill	50	784	
3	Corn	50	184	
4	Grass Pasture	50	447	
5	Grass Trees	50	697	
6	Hay Windrowed	50	439	
7	Soybean Notill	50	918	
8	Soybean Mintill	50	2418	
9	Soybean Clean	50	564	
10	Wheat	50	162	
11	Woods	50	1244	
12	Buildings Grass Trees Drives	50	330	
13	Stone Steel Towers	50	45	
14	Alfalfa	15	39	
15	Grass Pasture Mowed	15	11	
16	Oats	15	5	
	Total	695	9671	

TABLE II
Land-Cover Categories of the Augsburg Dataset and Its Corresponding Training and Testing Samples

No.	Category	Training	Testing	Color
1	Forest	146	13361	
2	Residential Area	264	30065	
3	Industrial Area	21	3830	
4	Low Plants	248	26609	
5	Allotment	52	523	
6	Commercial Area	7	1638	
7	Water	23	1507	
	Total	761	77533	

TABLE III
Land-Cover Categories of the Houston 2013 Dataset and Its Corresponding Training and Testing Samples

No.	Category	Training	Testing	Color
1	Healthy Grass	198	1053	
2	Stressed Grass	190	1064	
3	Synthetic Grass	192	505	
4	Tree	188	1056	
5	Soil	186	1056	
6	Water	182	143	
7	Residential	196	1072	
8	Commercial	191	1053	
9	Road	193	1059	
10	Highway	191	1036	
11	Railway	181	1054	
12	Parking Lot1	192	1041	
13	Parking Lot2	184	285	
14	Tennis Court	181	247	
15	Running Track	187	473	
	Total	2832	12197	

In this study, to quantitatively evaluate the performance of the proposed method, three widely used metrics are employed: overall accuracy (OA), average accuracy (AA), and kappa coefficient (κ).

B. Implementation Details

All experiments are conducted on a workstation equipped with an NVIDIA GeForce RTX 4090D GPU with 24 GB memory. The proposed FA3DHeat is implemented using the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:20:38 UTC from IEEE Xplore. Restrictions apply.

TABLE IV

COMPARISON WITH OTHER STATE-OF-THE-ART METHODS IN TERMS OF (OA%/AA%/κ) ON THE INDIAN PINES TEST SET

Category	CNN-based			Transformer-based			Mamba-based		Frequency-based	Heat Conduction-based	
	2-D CNN	3-D CNN	HybridSN	ViT	MorphFormer	SSFTT	ViM	S^2Mamba	MSFI-CNet	HCI-Net	**FA3DHeat**
1	72.59±6.84	77.13±5.62	79.85±4.97	73.85±7.15	92.47±3.26	94.14±2.41	75.80±6.70	88.28±5.18	**95.68±2.07**	90.93±3.32	92.82±2.18
2	75.54±5.91	97.00±1.74	89.93±3.86	85.25±4.52	**99.76±0.38**	93.88±2.96	93.88±3.48	97.60±1.26	98.80±0.82	98.68±0.93	99.64±0.41
3	98.29±1.08	95.30±2.53	97.44±1.76	93.16±3.44	100.00±0.00	99.57±0.51	98.72±1.13	99.57±0.46	98.72±0.97	99.57±0.58	**100.00±0.00**
4	86.92±4.37	95.17±2.49	93.56±3.12	94.16±2.86	98.56±0.91	95.77±2.37	93.16±3.15	97.59±1.28	98.59±0.78	**98.79±0.83**	98.59±0.62
5	94.78±2.86	89.83±4.71	94.11±2.92	93.44±3.05	**100.00±0.00**	98.66±0.94	96.12±2.24	**100.00±0.00**	99.87±0.31	99.46±0.52	98.66±1.04
6	98.16±1.37	96.52±2.11	99.39±0.74	97.75±1.46	**100.00±0.00**	**100.00±0.00**	91.82±4.29	99.59±0.53	**100.00±0.00**	**100.00±0.00**	99.80±0.28
7	86.36±5.26	85.43±6.18	94.11±3.51	83.99±6.72	97.42±1.48	88.74±5.64	87.71±5.37	90.90±4.82	**98.45±0.91**	89.77±4.16	96.69±1.35
8	78.44±6.58	76.58±7.24	82.74±5.91	70.46±8.65	97.24±1.36	88.53±4.79	78.77±6.96	93.51±3.24	92.18±3.38	94.49±2.75	**97.73±0.86**
9	87.95±4.91	84.85±5.46	87.13±4.88	84.36±6.02	93.81±3.13	92.02±3.51	82.90±6.34	93.16±3.26	**96.25±1.72**	93.32±2.94	96.09±1.21
10	99.53±0.62	99.53±0.58	100.00±0.00	99.06±0.94	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	**100.00±0.00**
11	86.94±5.18	97.37±1.65	95.83±2.38	84.47±6.71	98.53±1.14	97.30±1.46	94.28±3.05	**98.76±0.84**	98.45±0.91	98.38±1.03	98.68±0.73
12	91.05±4.12	95.79±2.34	95.79±2.19	82.89±7.52	98.95±0.86	97.63±1.75	76.32±8.41	98.68±0.92	99.47±0.54	97.37±1.68	**100.00±0.00**
13	100.00±0.00	100.00±0.00	100.00±0.00	98.95±0.88	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	**100.00±0.00**
14	35.19±22.84	94.44±2.16	94.44±3.28	98.15±1.09	100.00±0.00	100.00±0.00	96.29±1.43	100.00±0.00	100.00±0.00	96.30±2.21	**100.00±0.00**
15	100.00±0.00	100.00±0.00	100.00±0.00	96.15±2.94	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	**100.00±0.00**
16	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	**100.00±0.00**
OA (%)	83.69±4.82	87.01±3.27	89.46±2.91	82.11±4.95	97.40±1.18	93.71±2.16	86.32±4.24	95.04±1.53	96.76±0.94	95.55±1.41	**97.56±0.43**
AA (%)	86.98±3.76	92.81±2.48	94.02±2.17	89.76±4.18	98.56±0.74	96.64±1.35	91.61±3.61	97.35±0.96	98.53±0.61	97.32±1.08	**98.67±0.31**
κ (%)	81.52±5.31	85.33±3.72	88.07±3.25	79.84±5.48	97.04±1.29	92.85±2.37	84.54±4.76	94.35±1.70	96.32±1.06	94.94±1.56	**97.22±0.47**

TABLE V

COMPARISON WITH OTHER STATE-OF-THE-ART METHODS IN TERMS OF (OA%/AA%/κ) ON THE AUGSBURG TEST SET

Category	CNN-based			Transformer-based			Mamba-based		Frequency-based	Heat Conduction-based	
	2-D CNN	3-D CNN	HybridSN	ViT	MorphFormer	SSFTT	ViM	S^2Mamba	MSFI-CNet	HCI-Net	**FA3DHeat**
1	97.49±1.26	86.36±4.85	90.87±3.92	88.39±4.38	93.95±2.74	**98.16±0.84**	93.91±2.86	95.69±2.17	97.18±1.28	95.97±1.96	97.18±1.13
2	94.73±1.37	96.19±1.86	92.16±3.54	91.25±4.16	**97.38±1.42**	97.06±1.31	96.15±1.88	96.94±2.46	96.40±1.72	97.19±1.35	96.87±1.24
3	53.10±8.92	74.97±6.38	41.03±10.74	65.83±7.65	47.44±11.28	63.23±8.36	51.62±9.47	67.20±7.84	74.79±5.96	73.54±6.43	**78.78±4.26**
4	78.20±2.64	90.82±3.48	91.24±3.27	84.31±5.12	93.93±2.58	**94.82±2.16**	86.81±4.73	94.69±2.35	93.44±2.61	90.80±3.72	92.40±2.87
5	71.48±8.36	65.04±9.18	46.61±12.52	60.35±9.74	39.65±12.86	56.70±10.21	70.61±8.74	71.65±7.96	56.17±10.43	59.30±5.65	**81.04±4.82**
6	8.93±5.74	4.44±3.28	9.60±6.15	6.02±4.66	10.64±6.83	6.80±4.91	7.17±5.23	**12.89±7.26**	5.90±4.37	0.85±0.72	10.70±5.86
7	39.48±8.63	23.53±6.72	50.33±7.36	50.07±10.28	52.55±5.64	41.37±6.91	**57.52±3.46**	30.20±11.37	48.37±3.85	48.76±5.26	55.62±4.14
OA (%)	84.44±2.86	88.03±2.12	86.22±1.74	84.42±4.28	90.24±1.63	91.53±0.84	87.78±2.36	91.23±1.21	91.32±0.86	90.38±1.47	**91.77±0.72**
AA (%)	63.34±2.94	63.05±2.37	60.26±3.28	63.75±2.41	62.22±3.16	65.45±1.82	66.26±1.47	67.04±3.73	67.46±2.96	66.63±3.38	**73.23±1.26**
κ (%)	77.87±3.52	82.62±3.63	80.29±2.28	77.73±3.91	85.83±3.04	87.80±0.48	82.41±2.97	87.36±1.54	87.58±1.16	86.12±2.85	**88.24±0.76**

TABLE VI

COMPARISON WITH OTHER STATE-OF-THE-ART METHODS IN TERMS OF (OA%/AA%/κ) ON THE HOUSTON 2013 TEST SET

Category	CNN-based			Transformer-based			Mamba-based		Frequency-based	Heat Conduction-based	
	2-D CNN	3-D CNN	HybridSN	ViT	MorphFormer	SSFTT	ViM	S^2Mamba	MSFI-CNet	HCI-Net	**FA3DHeat**
1	86.49±5.26	85.21±4.82	85.69±6.57	86.57±4.13	84.89±5.06	85.37±4.68	**90.73±3.15**	85.21±3.76	85.21±2.38	85.53±4.19	85.69±2.84
2	98.80±0.74	98.33±0.96	96.41±1.85	98.96±0.62	**100.00±0.00**	97.53±1.26	97.05±1.52	99.52±0.36	99.36±0.47	**100.00±0.00**	99.12±0.54
3	95.55±2.17	98.42±0.83	98.28±1.91	99.28±0.42	**100.00±0.00**	**100.00±0.00**	99.86±0.21	99.57±0.38	99.86±0.18	99.71±0.29	99.57±0.35
4	98.87±0.66	**99.36±0.41**	97.27±1.34	98.15±0.94	98.79±0.72	90.43±3.58	99.12±0.53	98.23±0.86	99.20±0.46	98.39±0.78	99.04±0.49
5	99.84±0.22	97.83±1.12	99.76±0.31	99.84±0.20	98.71±0.84	99.11±0.57	98.39±1.06	100.00±0.00	100.00±0.00	100.00±0.00	**100.00±0.00**
6	94.77±2.36	98.16±0.94	98.15±0.98	98.15±0.91	98.15±0.87	98.15±0.92	97.85±1.16	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	98.15±0.83
7	82.41±5.48	93.93±2.63	88.64±4.12	87.07±4.86	95.11±2.24	85.80±3.17	91.96±3.31	**95.66±1.86**	90.54±3.54	87.30±4.73	91.48±2.96
8	84.32±5.06	80.79±3.31	77.33±6.84	80.31±6.56	85.37±4.72	84.97±2.85	85.53±4.36	92.28±2.57	92.68±2.18	**93.81±1.74**	92.28±2.03
9	77.55±6.72	87.30±4.38	80.19±4.16	80.19±6.04	90.58±3.42	90.10±3.76	84.98±5.12	88.26±4.08	90.34±3.15	88.42±3.96	**93.29±1.63**
10	65.53±8.94	61.94±4.67	55.50±10.86	65.53±8.52	76.69±6.34	77.51±3.02	51.26±11.42	86.39±4.26	87.45±5.74	86.96±4.05	**97.39±1.12**
11	89.39±4.25	76.03±7.68	87.85±4.72	77.98±7.24	**99.84±0.25**	87.69±4.88	86.07±5.36	99.27±0.54	99.27±0.57	91.09±3.42	97.17±1.28
12	92.38±3.76	82.89±6.14	**94.73±2.58**	80.86±6.83	93.27±3.12	91.57±3.86	84.51±5.75	85.08±5.46	93.67±2.72	83.05±6.21	91.08±2.41
13	94.03±2.84	93.60±3.12	88.70±4.96	**95.10±2.31**	89.55±4.43	86.78±4.24	93.18±3.06	89.98±4.15	90.62±3.84	93.18±2.95	93.39±3.04
14	98.36±0.93	86.21±5.48	99.53±0.38	99.77±0.26	100.00±0.00	100.00±0.00	99.07±0.74	100.00±0.00	100.00±0.00	100.00±0.00	**100.00±0.00**
15	91.36±3.92	98.94±0.72	98.03±1.18	98.79±0.86	100.00±0.00	98.64±0.95	99.39±0.46	100.00±0.00	100.00±0.00	100.00±0.00	**100.00±0.00**
OA (%)	88.78±2.84	88.00±1.26	88.14±3.04	87.78±3.31	93.29±1.62	90.39±2.53	88.93±2.91	93.87±1.24	94.54±0.76	92.70±1.58	**95.30±0.37**
AA (%)	89.98±2.36	89.26±1.58	89.74±2.64	89.77±2.91	94.06±1.37	91.58±2.16	90.60±2.48	94.63±1.06	95.21±0.62	93.83±1.31	**95.84±0.31**
κ (%)	87.87±3.08	87.03±1.55	87.19±3.31	86.79±3.62	92.75±1.78	89.63±2.76	88.04±3.18	93.38±1.37	94.09±0.84	92.11±1.74	**94.91±0.42**

PyTorch framework. For fair comparison, all competing methods follow the same training and testing split protocol as S^2Mamba [48], where a small number of labeled samples are selected for training, and the remaining labeled samples are used for testing. Under the fixed training and testing splits, each method is repeated ten times with different random seeds, and the mean and standard deviation are reported to evaluate performance stability. The hyperparameters of competing methods are set according to the corresponding original papers or official implementations whenever available.

For all datasets, FA3DHeat is optimized using the Adam optimizer with a mini-batch size of 64. The initial learning rate is set to 1×10^{-3} and decayed by a factor of 0.9 every 50 epochs. The network is trained for 500 epochs on the Indian Pines and Augsburg datasets, and for 400 epochs on the Houston 2013 dataset. The input patch size is set to 7×7 for Indian Pines and Augsburg, and 9×9 for Houston 2013. All input samples are normalized to the range $[0, 1]$ using band-wise min–max normalization.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:20:38 UTC from IEEE Xplore. Restrictions apply.

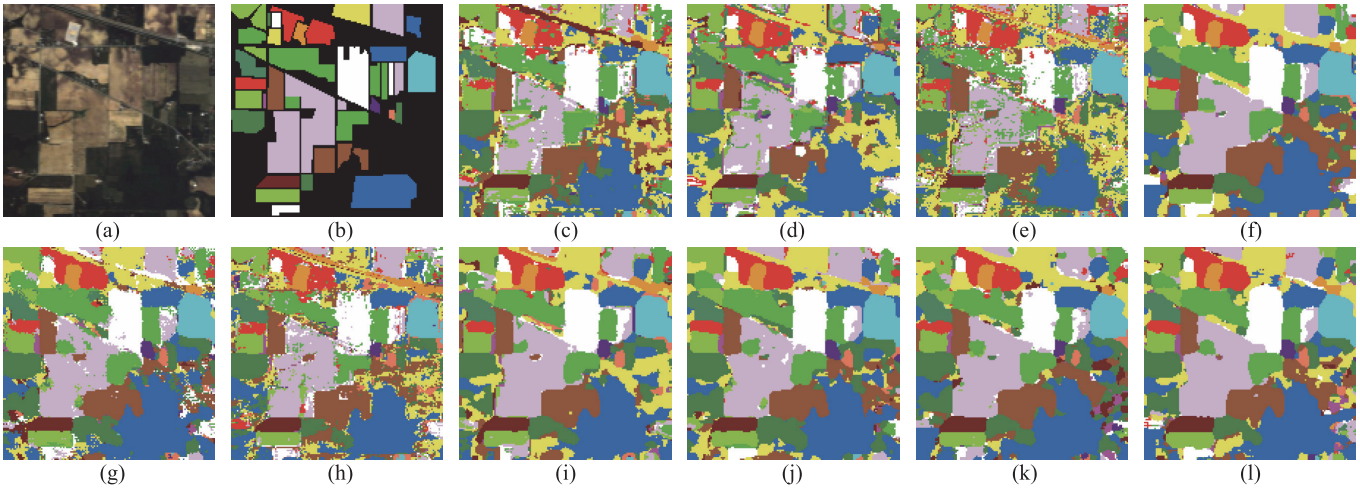


Fig. 4. Qualitative classification results on the Indian Pines dataset. (a) False-color map, (b) Ground truth, (c) 2-D CNN [9], (d) 3-D CNN [10], (e) ViT [49], (f) MorphFormer [50], (g) SSFTT [12], (h) ViM [51], (i) S^2Mamba [48], (j) MSFI-CNet [52], (k) HCI-Net [21], and (l) FA3DHeat.

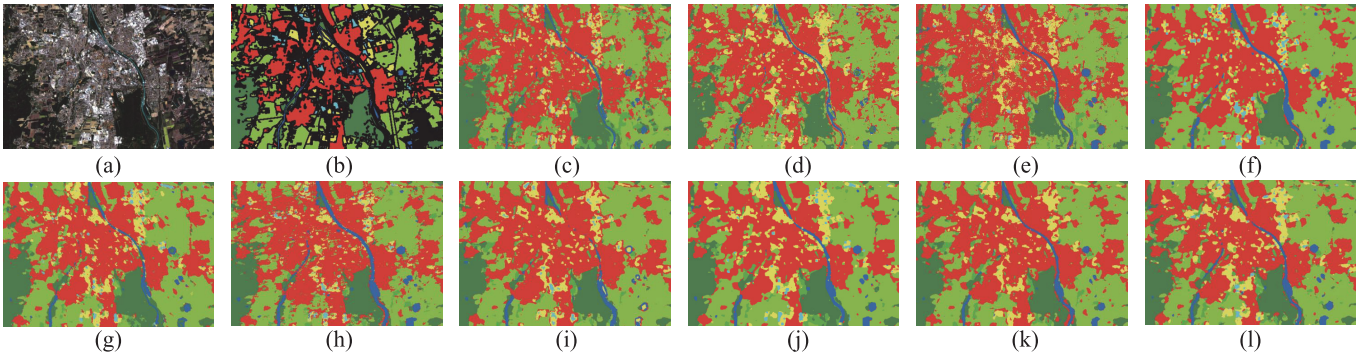


Fig. 5. Qualitative classification results on the Augsburg dataset. (a) False-color map, (b) Ground truth, (c) 2-D CNN [9], (d) 3-D CNN [10], (e) ViT [49], (f) MorphFormer [50], (g) SSFTT [12], (h) ViM [51], (i) S^2Mamba [48], (j) MSFI-CNet [52], (k) HCI-Net [21], and (l) FA3DHeat.

C. Comparison With State-of-the-Art Methods

To validate the classification performance of the proposed FA3DHeat, we compare it with several representative methods from different modeling paradigms. The compared methods include CNN-based models, i.e., 2-D CNN [9], 3-D CNN [10], and HybridSN [11]; Transformer-based models, i.e., ViT [49], MorphFormer [50], and SSFTT [12]; Mamba-based models, i.e., Vision Mamba (ViM) [51] and S^2Mamba [48]; a frequency-based model, i.e., MSFI-CNet [52]; and a heat conduction-based model, i.e., HCI-Net [21]. These baselines cover local convolutional feature extraction, token-based global dependency modeling, sequential state-space modeling, frequency-domain representation learning, and heat-based feature propagation.

Tables IV–VI report the class-wise accuracies and overall quantitative results in terms of OA, AA, and κ. The best result for each row is highlighted in bold. Figs. 4–6 present representative classification maps for qualitative comparison.

1) Indian Pines: As shown in Table IV, FA3DHeat obtains the highest OA, AA, and κ among all compared methods on the Indian Pines dataset. Compared with MorphFormer, which provides the strongest competing overall results, FA3DHeat achieves marginally higher OA, AA, and κ by 0.16, 0.11, and 0.18 percentage points, respectively. Meanwhile, FA3DHeat records the lowest standard deviations of 0.43, 0.31, and 0.47 for OA, AA, and κ, respectively, indicating reduced performance variation across repeated runs. Compared with CNN-based methods, FA3DHeat obtains clear performance gains, indicating that 3-D heat conduction is more effective than purely local convolutional extraction for modeling coupled spectral–spatial information. Compared with Transformer- and Mamba-based methods, FA3DHeat achieves competitive performance, suggesting that frequency-domain 3-D heat propagation can effectively capture discriminative spectral–spatial dependencies without relying on dense self-attention or sequential scanning.

In terms of class-wise accuracy, FA3DHeat performs well on several crop categories with similar spectral responses, such as Corn-notill, Corn-mintill, Soybean-notill, and Soybean-mintill. This demonstrates the benefit of combining 3DHCO with the MBFA module, where anisotropic 3-D propagation strengthens spectral–spatial interaction and frequency-aware branches capture complementary low-, mid-, and high-frequency cues. The visual results in Fig. 4 further show that FA3DHeat produces more homogeneous classification regions

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:20:38 UTC from IEEE Xplore. Restrictions apply.

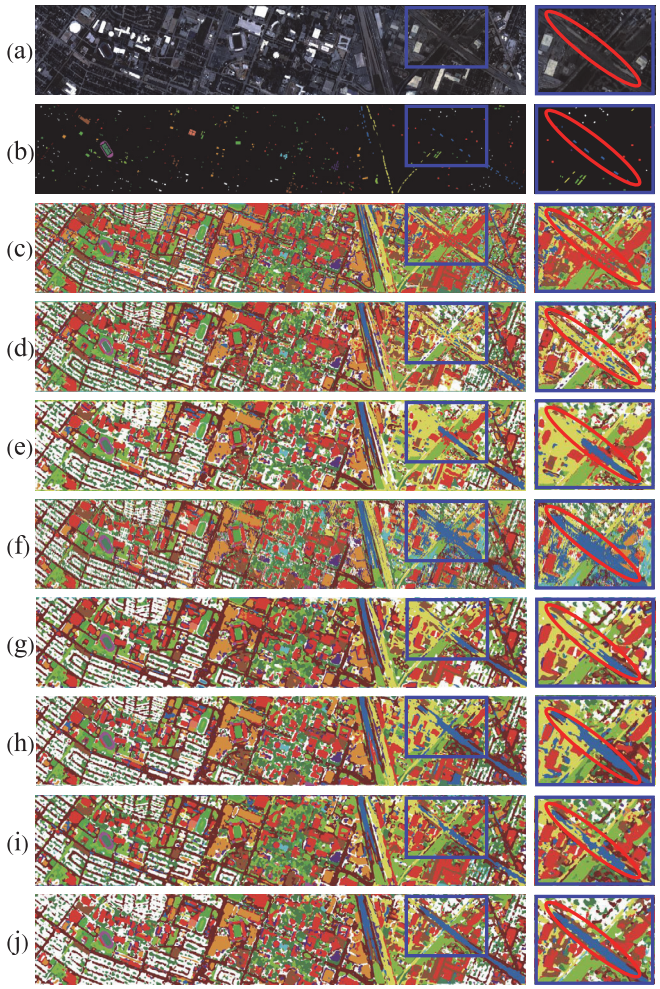


Fig. 6. Qualitative classification results on the Houston 2013 dataset. (a) False-color map, (b) Ground truth, (c) 2-D CNN [9], (d) 3-D CNN [10], (e) MorphFormer [50], (f) SSFTT [12], (g) S^2Mamba [48], (h) MSFI-CNet [52], (i) HCI-Net [21], and (j) FA3DHeat.

with fewer noisy predictions, which is consistent with the quantitative improvements in Table IV.

2) Augsburg: The Augsburg dataset is challenging due to its large spatial extent and complex land-cover distributions. As reported in Table V, FA3DHeat obtains the highest values for all three overall evaluation metrics among the compared methods, with an OA of 91.77%, an AA of 73.23%, and a κ of 88.24%. Compared with SSFTT, which obtains the highest competing OA and κ, FA3DHeat improves OA and κ by 0.24% and 0.44%, respectively. In terms of AA, FA3DHeat surpasses MSFI-CNet, the strongest competing method on this metric, by 5.77%.

It is worth noting that FA3DHeat achieves relatively high accuracies on several difficult categories, especially Categories 3 and 5, whose accuracies are relatively low for most competing methods. The relatively low accuracy of Category 6 can be mainly attributed to severe class imbalance, as only seven training samples are available, compared with 264 samples in the largest category, making it difficult to adequately capture its intraclass spectral–spatial variability. Nevertheless, the proposed frequency-aware 3-D heat conduction framework improves the discrimination of challenging land-cover categories in complex urban scenes. The improvement mainly benefits from the joint modeling of spectral and spatial dimensions in 3DHCO and the complementary frequency representations learned by the MBFA module. As further illustrated in Fig. 5, FA3DHeat yields cleaner classification maps with reduced noise while preserving spatially coherent regions and clearer class boundaries.

3) Houston 2013: Furthermore, we evaluate the proposed method on the Houston 2013 dataset, which contains complex urban structures and challenging class distributions. As reported in Table VI, FA3DHeat obtains the highest OA, AA, and κ among all compared methods, with an OA of 95.30%, an AA of 95.84%, and a κ of 94.91%. Compared with MSFI-CNet, which provides the strongest competing overall results, FA3DHeat improves OA, AA, and κ by 0.76%, 0.63%, and 0.82%, respectively. FA3DHeat also outperforms HCI-Net, suggesting that extending heat conduction from 2-D spatial feature propagation to 3-D spectral–spatial propagation is beneficial for HSI classification.

Fig. 6 presents the qualitative classification results. Compared with the competing methods, FA3DHeat produces more spatially coherent classification maps and better preserves structural details. In particular, the proposed method more accurately identifies the highway regions highlighted by the red ellipse, indicating that frequency-aware 3-D heat conduction is beneficial for enhancing structural continuity and reducing misclassification in complex urban areas.

D. Ablation Study

1) Ablation on Heat Conduction Operators: To investigate the effects of different heat conduction operators, we compare full 3DHCO with no conduction (implemented by removing heat decay, i.e., setting the heat attenuation term $\mathbf{A}$ to an all-one tensor), spectral-only HCO, spatial-only HCO, and a parameter-matched 3-D convolution. As reported in Table VII, full 3DHCO achieves higher OA, AA, and κ than the other variants on all three datasets. Compared with no conduction, it improves OA by 2.69%, 2.10%, and 2.22% on Indian Pines, Augsburg, and Houston 2013, respectively, demonstrating the effectiveness of heat conduction for hyperspectral representation learning.

Both spectral-only HCO and spatial-only HCO obtain higher OA than no conduction across all datasets, indicating that heat conduction along either the spectral or spatial dimension is beneficial for hyperspectral representation learning. Full 3DHCO further improves OA over spectral-only HCO by 0.99%, 1.23%, and 1.52%, and over spatial-only HCO by 1.08%, 1.33%, and 1.36%, respectively. These results suggest that joint spectral–spatial propagation can better exploit the complementary information from the two domains. Moreover, with a comparable parameter scale, full 3DHCO outperforms 3-D Conv by 1.55%, 1.54%, and 1.57% in OA, showing that frequency-domain spectral–spatial propagation is more effective than parameter-matched local 3-D convolution for the considered datasets.

2) Effect of the Number of Frequency Branches: To evaluate the effectiveness of the multifrequency branch design in the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:20:38 UTC from IEEE Xplore. Restrictions apply.

TABLE VII
COMPARISON OF DIFFERENT HEAT CONDUCTION OPERATORS

Heat conduction operator	Indian Pines			Augsburg			Houston 2013		
	OA (%)	AA (%)	κ (%)	OA (%)	AA (%)	κ (%)	OA (%)	AA (%)	κ (%)
No conduction	94.87	97.19	94.31	89.67	69.84	86.03	93.08	93.83	92.57
Spectral-only HCO	96.57	97.71	96.30	90.54	70.81	86.87	93.78	94.56	93.35
Spatial-only HCO	96.48	97.69	96.21	90.44	69.43	86.70	93.94	94.62	93.51
3D Conv	96.01	97.40	95.54	90.23	68.45	86.27	93.73	94.46	93.30
Full 3DHCO	**97.56**	**98.67**	**97.22**	**91.77**	**73.23**	**88.24**	**95.30**	**95.84**	**94.91**

TABLE VIII
ABLATION STUDY ON THE NUMBER OF FREQUENCY BRANCHES IN THE MBFA MODULE

Branches	Configuration	Indian Pines			Augsburg			Houston 2013		
		OA (%)	AA (%)	κ (%)	OA (%)	AA (%)	κ (%)	OA (%)	AA (%)	κ (%)
1	Mid	96.34	98.10	95.84	90.14	70.09	85.87	92.85	93.81	92.27
2	Low & High	97.09	98.60	96.67	90.91	70.97	87.06	94.77	95.51	94.35
3	Low & Mid & High	**97.56**	**98.67**	**97.22**	**91.77**	**73.23**	**88.24**	**95.30**	**95.84**	**94.91**

MBFA module, we conduct an ablation study by varying the number of frequency branches while keeping the other network components and training settings unchanged. Three configurations are considered. For the single-branch setting, only the mid-frequency branch is retained as a neutral frequency modeling baseline. For the two-branch setting, the low- and high-frequency branches are used to capture coarse structural information and fine local details, respectively. The three-branch setting corresponds to the complete low–mid–high frequency structure.

As reported in Table VIII, the classification performance improves consistently as the number of frequency branches increases. Compared with the single-branch setting, the complete three-branch configuration improves OA by 1.22%, 1.63%, and 2.45% on the Indian Pines, Augsburg, and Houston 2013 datasets, respectively. Similar improvements can also be observed in terms of AA and κ. These results indicate that relying on a single frequency component is insufficient to fully characterize the complex spectral–spatial patterns in HSI data. By jointly modeling low-, mid-, and high-frequency representations, the proposed MBFA module can capture complementary information from different frequency ranges, leading to more discriminative hyperspectral representations.

3) Ablation on LSMs and AFBF: To analyze the contribution of the main components in the MBFA module, ablation experiments with different combinations of LSMs and the AFBF module are conducted on three datasets, as listed in Table IX. When LSMs are removed, the three branches are retained, but no branch-specific frequency selection is applied to the frequency-domain features. When AFBF is removed, the input-dependent channel-wise fusion strategy is replaced with fixed average fusion.

When AFBF is not adopted, introducing LSMs improves the OA by 0.27%, 0.58%, and 0.06% on Indian Pines, Augsburg, and Houston 2013, respectively. This indicates that LSM-based frequency selection helps different branches focus on suitable frequency ranges and enhances frequency-specific feature extraction. In addition, when LSMs are not adopted, introducing AFBF improves the OA by 0.43%, 0.79%, and 0.52% on the three datasets, respectively, demonstrating that adaptive branch aggregation is more effective than fixed average fusion.

TABLE IX
ABLATION STUDY ON LSMS AND AFBF IN THE MBFA MODULE

Module		Metrics	Datasets		
LSMs	AFBF		Indian Pines	Augsburg	Houston 2013
✗	✗	OA(%)	96.97	90.82	94.40
		AA(%)	98.19	70.27	95.25
		κ(%)	96.55	86.84	93.95
✓	✗	OA(%)	97.24	91.40	94.46
		AA(%)	98.51	70.83	95.24
		κ(%)	96.86	87.65	94.02
✗	✓	OA(%)	97.40	91.61	94.92
		AA(%)	98.56	71.57	95.67
		κ(%)	97.04	87.95	94.50
✓	✓	OA(%)	**97.56**	**91.77**	**95.30**
		AA(%)	**98.67**	**73.23**	**95.84**
		κ(%)	**97.22**	**88.24**	**94.91**

Furthermore, the simultaneous use of LSMs and AFBF achieves the best classification performance on all datasets. Compared with the setting without these two components, the OA is improved by 0.59%, 0.95%, and 0.90% on Indian Pines, Augsburg, and Houston 2013, respectively. These results show that LSMs and AFBF are complementary in the MBFA module. LSMs strengthen frequency-specific representation learning by guiding low-, mid-, and high-frequency branches toward different spectral–spatial frequency ranges, while AFBF adaptively adjusts their contributions according to input-dependent feature responses. Therefore, their joint use enables more effective frequency-aware spectral–spatial representation learning and improves classification performance in different HSI scenarios.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:20:38 UTC from IEEE Xplore. Restrictions apply.

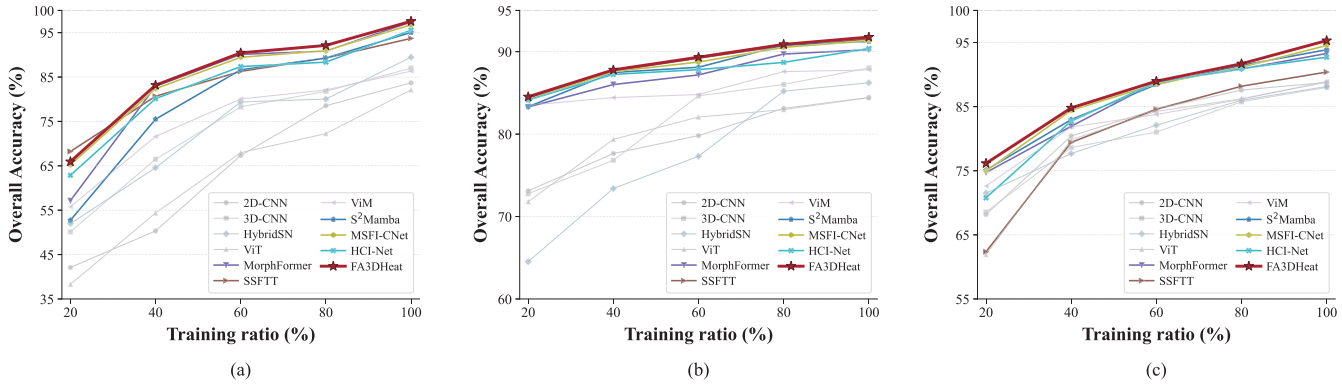


Fig. 7. Classification performance of representative methods under varying training ratios on three datasets. (a) Indian Pines. (b) Augsburg. (c) Houston 2013.

E. Performance Under Different Training Ratios

To further evaluate the performance of the proposed FA3DHeat under different training sample ratios, we compare it with representative competing methods using different proportions of training samples. Specifically, fixed training subsets corresponding to 20%, 40%, 60%, 80%, and 100% of the predefined training set are used for model training, while the testing set remains unchanged for all settings. OA is adopted as the evaluation metric. The corresponding results on the Indian Pines, Augsburg, and Houston 2013 datasets are shown in Fig. 7.

In general, the OA values of most methods increase as more training samples are available. On the Indian Pines dataset, FA3DHeat achieves competitive performance with limited training samples and obtains the best OA when the training ratio is increased to 40% and above. Although SSFTT performs slightly better at the 20% training ratio, FA3DHeat shows a more consistent improvement trend as the number of training samples increases.

On the Augsburg dataset, FA3DHeat consistently achieves the highest OA under all five training ratios, demonstrating its effectiveness under different training sample sizes. A similar trend can be observed on the Houston 2013 dataset, where FA3DHeat also ranks first across all training settings. In particular, under the low-sample settings of 20% and 40%, FA3DHeat achieves OA values of 76.15% and 84.80%, respectively. These results indicate that the proposed frequency-aware 3-D heat conduction framework can effectively learn discriminative spectral–spatial representations even when the available training samples are limited, leading to favorable generalization performance across different HSI scenarios.

F. Feature Visualization Analysis

To qualitatively evaluate the representation enhancement brought by the proposed frequency-aware modeling, we extract features from the same set of test samples on the Indian Pines dataset before and after the MBFA module, and visualize them using t-SNE [53]. As shown in Fig. 8, the features before the MBFA module exhibit obvious interclass overlap, where samples from different categories are mixed in the embedded space. This indicates that the representation produced before multifrequency branch modeling is still insufficiently discriminative.

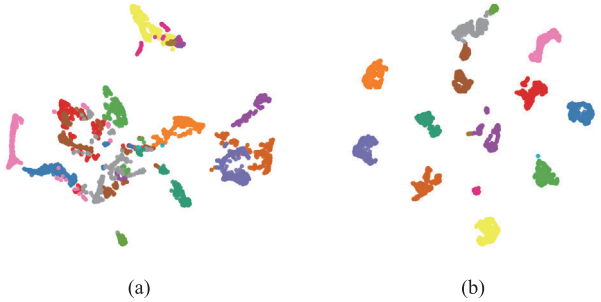


Fig. 8. t-SNE visualization of features on the Indian Pines dataset. (a) Features before the MBFA module. (b) Features after the MBFA module.

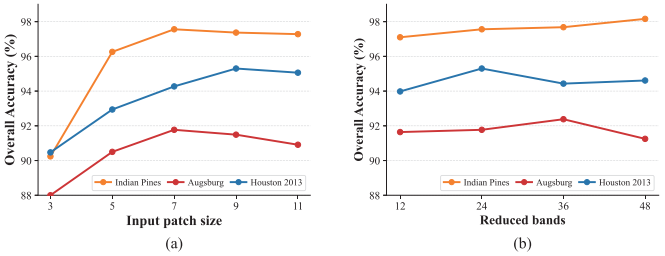


Fig. 9. Parameter analysis of FA3DHeat on three datasets. (a) Input patch size. (b) Number of reduced spectral bands.

After passing through the MBFA module, the embedded features form more compact intraclass clusters and clearer interclass separation. This improvement suggests that the MBFA module can effectively enhance feature discriminability by extracting complementary low-, mid-, and high-frequency spectral–spatial representations and adaptively integrating them through AFBF. The visualization results further demonstrate the effectiveness of frequency-aware 3-D heat conduction in learning more separable hyperspectral representations.

G. Parameter Analysis

In this section, we analyze the influence of two important hyperparameters in the proposed FA3DHeat, including the input patch size and the number of reduced spectral bands. During the analysis of each hyperparameter, the remaining settings are kept unchanged. The experiments are conducted on the Indian Pines, Augsburg, and Houston 2013 datasets, and OA is adopted as the evaluation metric.

Fig. 9(a) shows the classification performance under different input patch sizes. In general, increasing the patch size

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:20:38 UTC from IEEE Xplore. Restrictions apply.

TABLE X
COMPARISON OF RUNNING EFFICIENCY AND MODEL PARAMETERS FOR DIFFERENT METHODS

Methods	Running Time (s)						Params (M)
	Indian Pines		Augsburg		Houston 2013		
	Train	Test	Train	Test	Train	Test	
3-D CNN	223.94	1.65	765.30	13.20	723.69	83.38	1.22
ViT	229.15	2.38	1059.80	24.54	644.96	122.43	3.17
MorphFormer	210.52	1.99	930.91	19.86	500.94	93.01	0.28
MSFI-CNet	127.54	1.18	567.01	8.68	313.86	44.49	0.40
HCI-Net	185.60	1.47	807.20	7.76	420.52	49.34	0.54
FA3DHeat	195.63	1.87	866.74	20.39	625.87	124.85	0.49

provides more spatial contextual information and improves classification performance. However, an excessively large patch may introduce irrelevant neighboring pixels and increase computational cost. For Indian Pines and Augsburg, the best performance is achieved when the patch size is set to 7×7. For Houston 2013, a larger patch size of 9×9 leads to the highest OA, which may be attributed to its more complex urban structures and larger spatial regions. Therefore, the patch size is set to 7×7 for Indian Pines and Augsburg, and 9×9 for Houston 2013 in the experiments.

Fig. 9(b) illustrates the effect of the number of reduced spectral bands. A small number of reduced bands may discard useful spectral information, while an excessively large number of bands increases the computational burden of 3D-DCT/IDCT-based spectral–spatial propagation and may introduce redundant spectral responses. As observed, the performance of FA3DHeat remains relatively stable when the number of reduced bands changes from 24 to 48. Although larger reduced dimensions can bring slight improvements on some datasets, setting the number of reduced bands to 24 achieves competitive performance with lower computational cost. Therefore, we adopt 24 reduced bands as the default setting in the proposed method.

H. Comparison of Running Time and Model Parameters

To further evaluate the computational efficiency of the proposed FA3DHeat, we compare its parameter scale, training time, and inference time with several representative HSI classification methods on the Indian Pines, Augsburg, and Houston 2013 datasets.

As shown in Table X, FA3DHeat contains 0.49 M parameters, which is smaller than 3-D CNN, ViT, and HCI-Net, and slightly larger than MorphFormer and MSFI-CNet. This indicates that the proposed frequency-aware 3-D heat conduction framework maintains a compact parameter scale despite introducing 3DHCO, LSM-based frequency selection, and AFBF.

In terms of running time, FA3DHeat is not the fastest method, especially on the larger Houston 2013 dataset. This is mainly due to the 3D-DCT/IDCT operations used for spectral–spatial heat conduction and the parallel low-, mid-, and high-frequency branches in the MBFA module. Nevertheless, the overall computational cost remains within an acceptable range considering its compact model size and spectral–spatial representation capability. These results show that FA3DHeat achieves a reasonable tradeoff between model complexity and computational efficiency for HSI classification.

V. CONCLUSION

In this study, we propose FA3DHeat, a frequency-aware 3-D heat conduction network for HSI classification. The proposed method extends heat conduction from the 2-D spatial domain to the intrinsic 3-D spectral–spatial domain of HSI cubes. With the proposed 3DHCO, hyperspectral features can be effectively propagated and aggregated along both spectral and spatial dimensions. Furthermore, an MBFA module is designed to capture complementary features from low-, mid-, and high-frequency components. An AFBF module is further introduced to dynamically integrate these frequency-specific features according to input-dependent spectral–spatial statistics. Extensive experiments on three public HSI classification datasets demonstrate that FA3DHeat achieves competitive performance compared with several state-of-the-art methods. In future work, we will explore more efficient heat conduction operators and extend the proposed framework to other HSI interpretation tasks.

REFERENCES

[1] M. E. Schaepman, S. L. Ustin, A. J. Plaza, T. H. Painter, J. Verrelst, and S. Liang, "Earth system science related imaging spectroscopy—An assessment," *Remote Sens. Environ.*, vol. 113, pp. S123–S137, Sep. 2009.

[2] Y. Bai et al., "Hyperspectral remote sensing for monitoring crop disease: Applications, challenges, and perspectives," *IEEE Geosci. Remote Sens. Mag.*, vol. 14, no. 1, pp. 96–120, Feb. 2026.

[3] D. A. Abuhani, I. Zualkernan, R. Aldamani, and M. Alshafai, "Generative artificial intelligence for hyperspectral sensor data: A review," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 6422–6439, 2025.

[4] M. Sheykhmousa, M. Mahdianpari, H. Ghanbari, F. Mohammadimanesh, P. Ghamisi, and S. Homayouni, "Support vector machine versus random forest for remote sensing image classification: A meta-analysis and systematic review," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 6308–6325, 2020.

[5] J. Xia, N. Falco, J. A. Benediktsson, P. Du, and J. Chanussot, "Hyperspectral image classification with rotation random forest via KPCA," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 4, pp. 1601–1609, Apr. 2017.

[6] Y. Chen et al., "Unsupervised Bayesian subpixel mapping of hyperspectral imagery based on band-weighted discrete spectral mixture model and Markov random field," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 1, pp. 162–166, Jan. 2021.

[7] W. Diao, L. Hu, K. Zhang, and L. Xiao, "Dual-conditionally guided diffusion models for fusion of unregistered multisource remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5404117.

[8] V. Kumar, R. S. Singh, M. Rambabu, and Y. Dua, "Deep learning for hyperspectral image classification: A survey," *Comput. Sci. Rev.*, vol. 53, Aug. 2024, Art. no. 100658.

[9] Y. Chen, H. Jiang, C. Li, X. Jia, and P. Ghamisi, "Deep feature extraction and classification of hyperspectral images based on convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 6232–6251, Oct. 2016.

[10] Z. Zhong, J. Li, Z. Luo, and M. Chapman, "Spectral–spatial residual network for hyperspectral image classification: A 3-D deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 847–858, Feb. 2018.

[11] S. K. Roy, G. Krishna, S. R. Dubey, and B. B. Chaudhuri, "HybridSN: Exploring 3-D–2-D CNN feature hierarchy for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 2, pp. 277–281, Feb. 2020.

[12] L. Sun, G. Zhao, Y. Zheng, and Z. Wu, "Spectral–spatial feature tokenization transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, Jan. 2022, Art. no. 5522214.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:20:38 UTC from IEEE Xplore. Restrictions apply.

[13] H. Yu, Z. Xu, K. Zheng, D. Hong, H. Yang, and M. Song, "MSTNet: A multilevel spectral–spatial transformer network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5532513.

[14] Q. Liu, J. Yue, Y. Fang, S. Xia, and L. Fang, "HyperMamba: A spectral–spatial adaptive mamba for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5536514.

[15] Y. He, B. Tu, B. Liu, J. Li, and A. Plaza, "3DSS-Mamba: 3D-spectral–spatial mamba for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5534216.

[16] A. Yang, M. Li, Y. Ding, L. Fang, Y. Cai, and Y. He, "GraphMamba: An efficient graph structure learning vision mamba for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5537414.

[17] J. Zhang, C. Zhang, S. Liu, Z. Shi, and B. Pan, "Three-dimensional frequency-domain transform network for cross-scene hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5409813.

[18] M. Ahmad, U. Ghous, M. Usama, and M. Mazzara, "WaveFormer: Spectral–spatial wavelet transformer for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[19] Z. Wang, Y. Liu, Y. Tian, Y. Liu, Y. Wang, and Q. Ye, "Building vision models upon heat conduction," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2025, pp. 9707–9717.

[20] H. Hu et al., "RS-vHeat: Heat conduction guided efficient remote sensing foundation model," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2025, pp. 9876–9887.

[21] L. Liang et al., "HCI-Net: Heat conduction interaction network for hyperspectral image classification," *IEEE Trans. Circuits Syst. Video Technol.*, early access, 2026, doi: 10.1109/TCSVT.2026.3722712.

[22] C. Shi, D. Liao, T. Zhang, and L. Wang, "Hyperspectral image classification based on expansion convolution network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5528316.

[23] X. Zhang, S. Shang, X. Tang, J. Feng, and L. Jiao, "Spectral partitioning residual network with spatial attention mechanism for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022.

[24] S.-Y. Chen, P.-Y. Chu, K.-L. Liu, and Y.-C. Wu, "A multichannel hybrid 2D–3D-CNN for hyperspectral image classification with small training sample sizes," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5540915.

[25] G. Sun et al., "Large kernel spectral and spatial attention networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5519915.

[26] C. Yu, Y. Zhu, M. Song, Y. Wang, and Q. Zhang, "Unseen feature extraction: Spatial mapping expansion with spectral compression network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5521915.

[27] X. Liu, A. H.-M. Ng, F. Lei, J. Ren, L. Guo, and Z. Du, "MSLKCNN: A simple and powerful multiscale large kernel CNN for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5513314.

[28] P. Zhang, C. Wu, D. Wang, Z. Bai, T. Li, and Y. Li, "HC-XLSTM: A dual-branch framework with paired blocks for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 19, pp. 5967–5983, 2026.

[29] D. Hong et al., "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5518615.

[30] S. Mei, C. Song, M. Ma, and F. Xu, "Hyperspectral image classification using group-aware hierarchical transformer," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5539014.

[31] S. Zhang, J. Zhang, X. Wang, J. Wang, and Z. Wu, "ELS2T: Efficient lightweight spectral–spatial transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5518416.

[32] Z. Zhao, X. Xu, S. Li, and A. Plaza, "Hyperspectral image classification using groupwise separable convolutional vision transformer network," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5511817.

[33] S. Cheng, R. Chan, and A. Du, "MS2I2Former: Multiscale spatial–spectral information interactive transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5532919.

[34] J. Zou, W. He, and H. Zhang, "PSFormer: Pyramid superpixel transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5532816.

[35] A. Shafique, M. Siraj, S. Sohail, S. A. Alsaif, and B. Cheng, "AssociativeFormer: Integrating sparse memory attention and versatile tokenization for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 19, pp. 14247–14261, 2026.

[36] Y. Li, Y. Luo, L. Zhang, Z. Wang, and B. Du, "MambaHSI: Spatial–spectral mamba for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5524216.

[37] Y. He, B. Tu, P. Jiang, B. Liu, J. Li, and A. Plaza, "IGroupSS-Mamba: Interval group spatial–spectral mamba for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5538817.

[38] D. Li, U. A. Bhatti, M. Huang, L. Bruzzone, and J. Li, "HyPyraMamba: A pyramid spectral attention and mamba-based architecture for robust hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, 2026, Art. no. 5501016.

[39] J. Liu, L. Lian, Y. Li, and Y. Liu, "Hyperspectral image classification based on fractional Fourier transform," *Remote Sens.*, vol. 17, no. 12, p. 2065, Jun. 2025.

[40] X. Qiao and W. Huang, "A dual frequency transformer network for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 10344–10358, 2023.

[41] J. Gong, F. Li, J. Wang, Z. Yang, and X. Ding, "A split-frequency filter network for hyperspectral image classification," *Remote Sens.*, vol. 15, no. 15, p. 3900, Aug. 2023.

[42] H. Shi, G. Cao, Y. Zhang, Z. Ge, Y. Liu, and D. Yang, "F^3Net: Fast Fourier filter network for hyperspectral image classification," *IEEE Trans. Instrum. Meas.*, vol. 72, pp. 1–18, 2023.

[43] Z. Zhang and X. Gong, "Efficient visual representation learning with heat conduction equation," in *Proc. 33rd Int. Joint Conf. Artif. Intell.*, Aug. 2024, pp. 2431–2439.

[44] D. V. Widder, *The Heat Equation*. New York, NY, USA: Academic, 1975.

[45] R. O. Green et al., "Imaging spectroscopy and the airborne visible/infrared imaging spectrometer (AVIRIS)," *Remote Sens. Environ.*, vol. 65, no. 3, pp. 227–248, 1998.

[46] A. Baumgartner, P. Gege, C. Köhler, K. Lenhard, and T. Schwarzmaier, "Characterisation methods for the hyperspectral sensor HySpex at DLR's calibration home base," *Proc. SPIE*, vol. 8533, pp. 371–378, Jan. 2012.

[47] C. Debes et al., "Hyperspectral and LiDAR data fusion: Outcome of the 2013 GRSS data fusion contest," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2405–2418, Jun. 2014.

[48] G. Wang, X. Zhang, Z. Peng, T. Zhang, and L. Jiao, "S^2Mamba: A spatial–spectral state space model for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5511413.

[49] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2021, pp. 1–22.

[50] S. K. Roy, A. Deria, C. Shah, J. M. Haut, Q. Du, and A. Plaza, "Spectral–spatial morphological attention transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5503615.

[51] L. Zhu, B. Liao, Q. Zhang, X. Wang, W. Liu, and X. Wang, "Vision Mamba: Efficient visual representation learning with bidirectional state space model," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2024, pp. 62429–62442.

[52] Q. Liu, S. Fan, W. Li, S. Wei, and Y. Jiang, "A collaborative network via multi-head sparse and high-low frequency interaction for hyperspectral image classification," *Neurocomputing*, vol. 625, Apr. 2025, Art. no. 129541.

[53] L. Van der Maaten and G. Hinton, "Visualizing data using t-SNE," *J. Mach. Learn. Res.*, vol. 9, no. 86, pp. 2579–2605, 2008.



Dong Shi received the B.E. degree from the School of Computer Engineering, Jinling Institute of Technology, Nanjing, China, in 2024. He is currently pursuing the M.E. degree in electronic information with the College of Information Science and Technology and Artificial Intelligence, Nanjing Forestry University, Nanjing.

His research interests include deep learning and remote sensing image processing.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:20:38 UTC from IEEE Xplore. Restrictions apply.



Zhikang Xiang (Member, IEEE) received the B.S. degree in information and computing science from Nanjing University of Agriculture, Nanjing, China, in 2015, and the Ph.D. degree in computer science from Nanjing University of Science and Technology, Nanjing, in 2022.

He is currently an Associate Professor with the Department of Computer Information and Network Security, Jiangsu Police Institute, Nanjing. His fields of interests include deep learning and remote sensing image processing.



Yu Shen (Member, IEEE) received the B.E. degree in computer science and the Ph.D. degree in computer science from Nanjing University of Science and Technology, Nanjing, China, in 2015 and 2022, respectively.

He is currently an Associate Professor with the College of Information Science and Technology and Artificial Intelligence, Nanjing Forestry University, Nanjing. His research interests include remote sensing image analysis and processing, and pattern recognition.



Liang Xiao (Senior Member, IEEE) received the B.S. degree in applied mathematics and the Ph.D. degree in computer science from Nanjing University of Science and Technology (NJUST), Nanjing, China, in 1999 and 2004, respectively.

From 2009 to 2010, he was a Post-Doctoral Fellow with Rensselaer Polytechnic Institute, Troy, NY, USA. Since 2014, he has been the Deputy Director of the Jiangsu Key Laboratory of Spectral Imaging Intelligent Perception, NJUST. He has served as the Second Director of the Key Laboratory of Intelligent Perception and Systems for High-Dimensional Information, Ministry of Education, NJUST, where he is currently a Full Professor with the School of Computer Science. His main research interests include inverse problems in image processing, computer vision and image understanding, pattern recognition, and remote sensing.



Qian Du (Fellow, IEEE) received the Ph.D. degree in electrical engineering from the University of Maryland, Baltimore County, Baltimore, MD, USA, in 2000.

She is currently a Bobby Shackouls Professor with the Department of Electrical and Computer Engineering, Mississippi State University, Starkville, MS, USA. Her research interests include hyperspectral remote sensing image analysis and applications, pattern classification, data compression, and neural networks.

Dr. Du is a fellow of the SPIE—International Society for Optics and Photonics. She was a recipient of the 2010 Best Reviewer Award from the IEEE Geoscience and Remote Sensing Society (GRSS). She was the Co-Chair of the Data Fusion Technical Committee of the IEEE GRSS from 2009 to 2013. She was the Chair of the Remote Sensing and Mapping Technical Committee of the International Association for Pattern Recognition from 2010 to 2014. She was the General Chair of the fourth IEEE GRSS Workshop on Hyperspectral Image and Signal Processing: Evolution in Remote Sensing held in Shanghai, China, in 2012. She served as an Associate Editor for IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING (JSTARS), *Journal of Applied Remote Sensing*, and IEEE SIGNAL PROCESSING LETTERS.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:20:38 UTC from IEEE Xplore. Restrictions apply.